Self-Supervised Feature Learning Based on Spectral Masking for Hyperspectral Image Classification

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***Abstract*— Deep learning has emerged as a powerful method for hyperspectral image (HSI) classification. However, a significant prerequisite for HSI classification using deep learning is enough labeled samples, which is both time-consuming and labor-intensive. Yet, labeled samples are essential for training deep learning models. This article proposes an HSI classification method based on the self-supervised learning of spectral masking (SSLSM). The method mainly includes two steps: self-supervised pretraining and fine-tuning. First, considering the rich spectral information of HSI, we propose masked spectral reconstruction as the pretext task. The unmasked data are input into the encoder and decoder sequentially, which are composed of a multilayer transformer, for feature learning of masked spectral reconstruction. Second, we use reference samples to fine-tune the network, and the encoder and decoder are innovatively cascaded for deep semantic feature extraction, which can further improve the ability of feature extraction in the downstream classification tasks. The experimental results show that, compared with other methods, the SSLSM obtains the highest classification accuracy of 96.52%, 97.03%, and 96.70% on the Indian Pines dataset, Pavia University dataset, and Yancheng Wetlands dataset, respectively. Our method can also be applied to other HSI datasets, and the codes will be available from https://github.com/CIRSM-GRoup/2023-TGRS-SSLSM.**

***Index Terms*— Hyperspectral image (HSI) classification, self-supervised feature learning, spectral masking.**

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I. Introduction

LAND cover classification has a pivotal role in the human exploration of our planet. Remote sensing can periodically obtain ground object information in a large area and has been widely used in large-scale classification [1], [2]. Hyperspectral remote sensing has the advantages of narrow and continuous bands, making it a preferred choice for the classification of complex ground objects [3], [4], [5]. Compared with panchromatic and multispectral remote sensing, hyperspectral remote sensing always has more bands and higher spectral resolution. The spectral range can cover from visible to mid-infrared, and the spectral resolution can reach 4 nm, which can depict the land cover more precisely [4], [6]. Therefore, hyperspectral images (HSIs) have significant advantages in land cover classification.

HSI classification is the process of assigning unique labels to each pixel [7] and can be roughly divided into supervised and unsupervised classifications. The difference between them is whether the labeled samples are used. Initially, supervised methods, such as support vector machines (SVMs) [8] and random forests (RFs) [9], are used for HSI classification. However, its spatial features are ignored, which have a significant contribution in depicting the shape and texture of ground objects. Therefore, spatial feature extraction methods such as Gabor filters [10], local binary patterns [11], and morphological profiles [12] were also introduced to HSI classification. In addition, the fusion of spectral and spatial features can further improve the discriminative ability of the classifier [13], [14]. However, the aforementioned methods rely on shallow features of HSI and do not pay attention to capturing complex, high-level features, resulting in a lack of advantages in classification performance.

Deep learning can extract deep semantic features of data and has been widely applied in the fields of speech recognition [15], computer vision [16], and natural language processing [17]. The adaptation of deep learning has made great progress in HSI classification [18]. Convolutional neural networks (CNNs), as basic and typical deep learning networks for image processing, have been used in HSI classification [19], [20]. Its variations, including 1D-CNN [21], 2D-CNN [22], [23], and 3D-CNN [19], [24] also achieved high classification accuracy. In addition, researchers have explored the capabilities of the dual-branch network [25] and the hierarchical network [26] in HSI classification. However,

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CNN is prone to gradient disappearance and explosion problems, and residual networks can better solve this problem through residual connections [27]. As a sequence model, a recurrent neural network (RNN) is capable of processing temporal information in the spectral bands of HSI [28]. By utilizing features extracted by CNN as the input of the RNN, the two models can be combined to provide improved performance for HSI classification [29]. In addition to that, the transformer model can capture the global information through its self-attention mechanism. Several researchers have explored the potential applications of transformer models in HSI classification, such as HSI-bidirectional encoder representations from transformer (BERT) [30], central attention network (CAN) [31], SpectralFormer [32], and hyperspectral image transformer (HiT) [33]. These networks can extract high-level features through multiple layers of nonlinear transformations, thus achieving more accurate HSI classification.

The above deep learning models typically have a large number of parameters, and labeled samples are essential for deep learning-based supervised classification of HSI. By leveraging labeled samples, deep learning models can better learn the differences and commonalities between ground samples, leading to improved classification accuracy, generalization ability, and training efficiency. However, acquiring a large and precise set of labeled samples can be time-consuming and laborious. Therefore, deep learning classification for HSI often faces the challenge of insufficient labeled samples. Recently, utilizing abundant unlabeled data for feature learning has become a focus, such as semisupervised learning [34], [35], few-shot learning [36], and transfer learning [37]. Semisupervised learning is based on the labeled data and leverages the distribution information of the unlabeled data to construct the classifier and improve the classification accuracy, such as graph convolutional networks (GCNs) [38], [39] and generative adversarial networks (GANs) [40]. Few-shot learning is a form of meta-learning applied in the supervised learning field. Its purpose is to train a model using a limited amount of base class data and enable the model to adapt well to a new class that is disjoint with the base class [41]. Few-shot learning can mitigate the difficulty of imbalanced class data due to sample shortage and improve generalization ability by introducing regularization terms [42]. Transfer learning involves pretraining a model in the source domain with abundant samples and subsequently fine-tuning it using limited labeled data in the target domain [37]. It relaxes the assumption of independent and identically distributed training and test data, enabling the domains or tasks involved in learning to follow different marginal or conditional probability distributions. In cross-domain scenarios, it is common to encounter situations where the source and target domains are similar but not identical, leading to poor generalization performance of models in the target domain. Domain adaptation, as a specialized field of transfer learning, addresses this issue by balancing the differences between the source and target domains through learning the shared representation between them [43], [44]. In addition, there have been studies exploring the effectiveness of domain generalization in HSI classification by applying models trained from multiple source domains to new and unseen datasets [45], [46]. Among them, some studies have found that linguistic modal knowledge can improve the cross-scene HSI classification [45]. Both transfer learning and few-shot learning require a large amount of labeled data in the pretraining stage, while the acquisition of pseudo-labels in semisupervised learning is heavily influenced by labeled data; thus, their performance in HSI classification still can be improved.

Self-supervised learning (SSL) is a recently developed deep learning method in the field of unsupervised learning. Its main concept is to train a deep learning network for feature learning using appropriate pretext tasks from large-scale unlabeled data. This not only addresses the problem of limited labeled data, but also helps to alleviate overfitting. Additionally, compared to traditional methods of random parameter initialization, SSL can force the model to learn hidden representations of the input data through prediction or reconstruction tasks. When applied to downstream tasks, such as classification, fine-tuning the model with a small amount of labeled data can lead to higher classification accuracy. HSI has abundant unlabeled data, and rich ground object information, making it well-suited for applying SSL to HSI pretraining tasks. The key to SSL is pretext tasks [47], which include image rotation [48], relative position prediction [49], and image restoration [50]. The difference between SSL and unsupervised learning lies in whether supervised training is performed for pretext tasks, while for supervised learning, it is whether ground-truth labels are used. Currently, SSL in HSI can be divided into contrastive SSL and generative SSL.

The core of the contrastive SSL method is to learn feature representations by maximizing agreement between differently augmented views of the same input, by a carefully designed loss function [51]. Contrastive SSL in HSI classification can be mainly divided into local contrastive SSL and global contrastive SSL. Local contrastive SSL usually uses data augmentation techniques, such as Gaussian noise [7] and enhanced ensemble module [52], to generate positive and negative sample pairs and then trains the model using a contrastive loss function. In addition, Qin et al. [53] implemented SSL by comparing the similarity of different spatial neighborhoods based on data augmentation. They are easy to use and effective, as they only require a small number of positive and negative sample pairs to train the model. However, one disadvantage is that they are very sensitive to the choice of data augmentation, and local contrast makes it difficult to obtain global information. Global contrastive SSL methods use the entire image as positive and negative samples to train the model, instead of using local blocks [34], [54]. The advantage of these methods is that they can obtain global information, thus improving classification performance. However, since the entire image needs to be processed, the computational cost of these methods is higher.

In contrast, the primary concept of generative SSL is to use self-generated data to facilitate unsupervised learning,

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thus extracting useful features and learning representations of the data, leading to enhanced performance in downstream classification tasks. Once a model can generate the input data, it signifies that the model has learned the key features of the initial input image. Generative SSL can be categorized into two groups: prediction-based methods and reconstruction-based methods. Prediction-based methods involve pretext tasks that assist the model in learning pertinent features in HSI, such as angle rotation prediction (RP) and mirror image prediction [55], [56]. These methods provide diverse learning signals, resulting in better generalization performance. However, they may have relatively computational costs, and their design and implementation require significant expertise and experience. On the other hand, reconstruction-based methods train SSL models by minimizing the difference between the reconstructed data and the input data [57], [58], [59]. A representative pretext task for reconstruction-based methods is the reconstruction of masks. For instance, spatial masking (SM) involves random masking of HSI based on a certain proportion and then reconstructing the masked data through an encoder and decoder, achieving self-supervised feature learning [52], [54]. This process enables the learning of deep semantic feature representations, which helps to better understand the rich information contained in HSI.

Although the above self-supervised models have made some progress, there are still challenges in the field of HSI classification. First, existing self-supervised pretext tasks are either too simple or too complicated for HSI classification. Complex pretext tasks, such as SM [58], are prone to the loss of spatial and spectral information at the same time, which is detrimental to the reconstruction of pretrained models. Simple tasks, such as RP [55], indicate that the pretrained model has limited learning capacity, and thus, the classification accuracy is not satisfied. Second, models based on autoencoders do not fully take the advantage of feature extraction by the decoder. Therefore, in this article, SSL based on the spectral masking method is proposed for HSI classification. The main contributions are summarized as follows.

1) A new SSL based on spectral masking for HSI classification called self-supervised learning of spectral masking (SSLSM) is proposed. SSLSM fully considers the advantages of the high dimensionality of HSI data, includes a suitable pretext task for HSI classification, and can effectively utilize unlabeled samples.
2) Spectral masking is proposed as the pretext task of an SSL method for HSI classification. It preserves spatial information with a limited spectral mask. In addition, the encoder–decoder is cascaded in the fine-tuning stage, in which the decoder is used to further extract deep semantic features that cannot be extracted by the encoder alone. This is highly beneficial for downstream classification tasks.
3) Extensive experimental results demonstrate that SSLSM achieved optimal performance in land cover classification and showed strong robustness across three HSI datasets.

The remainder of this article is organized as follows. Section II provides a detailed explanation of our proposed self-supervised HSI classification method, SSLSM. Section III introduces the experimental datasets and analyzes and discusses the experimental results. Section IV concludes this article.

II. Methodology

A. Overview

Our method consists of two stages, pretraining and fine-tuning. In the pretraining stage, spectral masking reconstruction is proposed as the pretraining task. During this stage, the encoder maps the input bands that are not masked into a latent representation, and then, the decoder reconstructs the original bands based on latent representations and mask tokens. The loss between the reconstructed masked data and the original masked data is optimized to improve the feature extraction ability of the pretraining model. In the fine-tuning stage, the model cascades the encoder and decoder to extract semantic information and performs parameter fine-tuning based on a small number of labeled samples to achieve better HSI classification performance. Fig. 1 demonstrates the flowchart of SSLSM. The difference between the pretraining stage and fine-tuning stage is that the input HSI patch does not need to be masked and is directly input into the model for feature extraction, which can avoid information loss in the masked data.

B. Self-Supervised Pretraining

1) Spectral Masking: The original HSI data are denoted as $\boldsymbol{X} \in \mathbb{R}^{m\times n\times l}$, where $m \times n$ is the spatial size, and l is the number of bands. Typically, HSI can have hundreds of bands, which bring rich spectral information and dimensionality disaster at the same time. Therefore, principal component analysis (PCA) is first applied to reduce the original l bands to d bands, while keeping the spatial dimension unchanged. After PCA, the HSI can be represented as $\boldsymbol{X}_{\text{pca}} \in \mathbb{R}^{m\times n\times d}$. After that, small patches are extracted from the original HSI with the size of $s \times s \times d$, and each patch can be represented as $\boldsymbol{P} \in \mathbb{R}^{s\times s\times d}$, where $s \times s$ is the size of the sliding window. After patch generation, a certain proportion of bands are randomly masked and it is flattened into a 1-D vector through unfold; the specific operation is as follows:

$$\boldsymbol{P}_{\text{u_m}}, \boldsymbol{P}_{\text{m}} = \text{Unfold}(\text{Mask}(\boldsymbol{P})) \tag{1}$$

where $\boldsymbol{P}_{\text{u_m}} \in \mathbb{R}^{h\times s^2}$ and $\boldsymbol{P}_{\text{m}} \in \mathbb{R}^{z\times s^2}$ represent the unmasked and masked data, respectively. s^2 represents $s \times s$, and h and z represent the number of remaining bands after masking and the masked bands. Before $\boldsymbol{P}_{\text{u_m}}$ is input to the encoder, position encodings are added to provide position information

$$\boldsymbol{P}^0_{\text{u_m}} \leftarrow \boldsymbol{P}_{\text{u_m}} + \boldsymbol{E}_{\text{u_m}} \tag{2}$$

where $\boldsymbol{E}_{\text{u_m}} \in \mathbb{R}^{h\times s^2}$ is the position encoding information.

2) Encoder: After the position encoding embedding operation, it is input into the encoder (T_{en}) for feature learning and to obtain the latent representation; the formula is as follows:

$$\boldsymbol{G} = T_{\text{en}}(\boldsymbol{P}^0_{\text{u_m}}) \tag{3}$$

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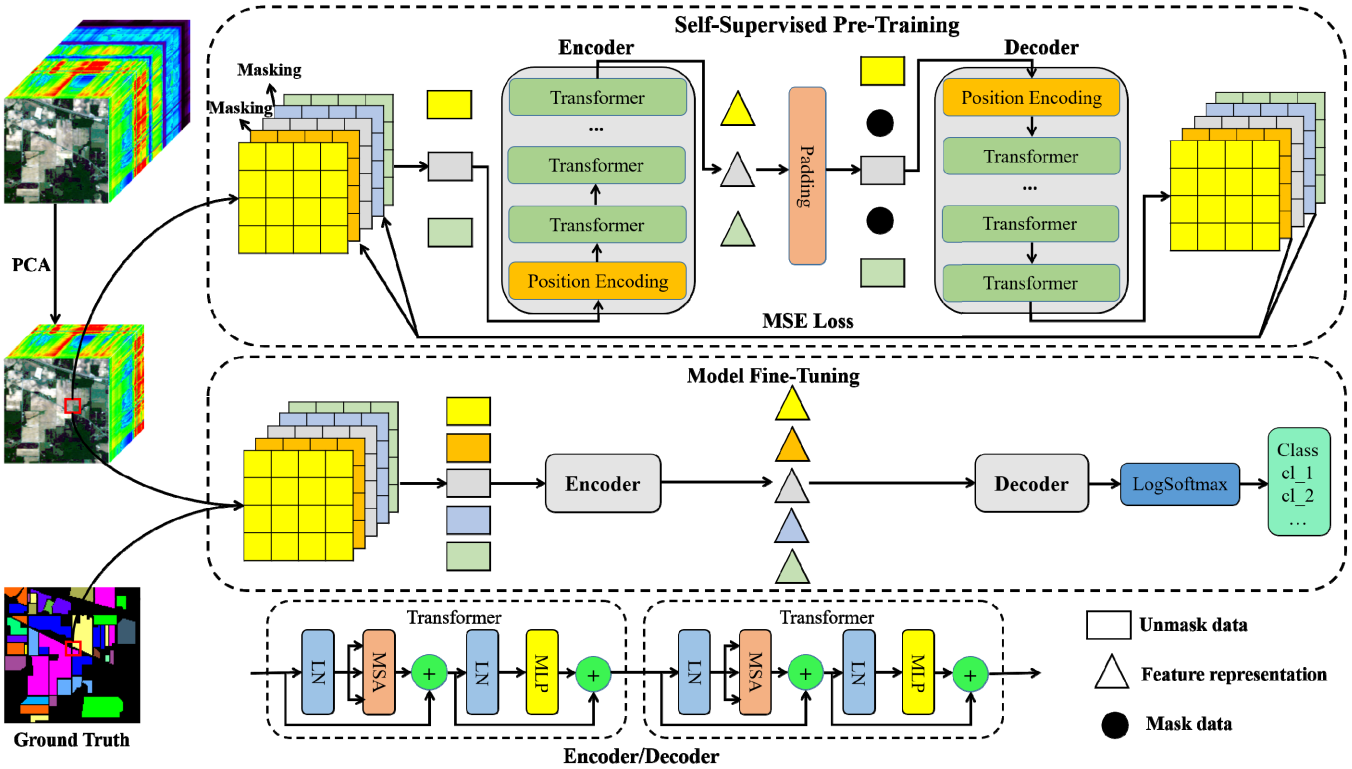

Fig. 1. Flowchart of the proposed SSLSM method.

where $\boldsymbol{G} \in \mathbb{R}^{h\times s^2}$, and the specific transformer operation is as follows:

$$\boldsymbol{P}^{v}_{\mathrm{u_m'}} = \boldsymbol{P}^{v-1}_{\mathrm{u_m}} + \mathrm{MSA}(\mathrm{LN}(\boldsymbol{P}^{v-1}_{\mathrm{u_m}})) \tag{4}$$

$$\boldsymbol{G} = \boldsymbol{P}^{v}_{\mathrm{u_m'}} + \mathrm{MLP}(\mathrm{LN}(\boldsymbol{P}^{v}_{\mathrm{u_m'}})) \tag{5}$$

where $v = 1, 2, \ldots, V$ represents the vth transformer layer, V represents the total number of transformer layers in the encoder, and $\boldsymbol{P}^{v}_{\mathrm{u_m'}} \in \mathbb{R}^{h\times s^2}$ is the intermediate output of the vth transformer layer in the encoder after layer normalization (LN) and multihead self-attention (MSA). LN layer can not only reduce computation time, but also alleviate the problem of gradient disappearance or explosion. MLP represents multilayer perceptron, which is used for feature and linear transformation between two self-attention layers. MSA is used to capture the relevant importance of the input sequence data. Input $\boldsymbol{P}^{0}_{\mathrm{u_m}} \in \mathbb{R}^{h\times s^2}$ is linearly transformed into three parts, query $Q \in \mathbb{R}^{s^2\times h_k}$, key $K \in \mathbb{R}^{s^2\times h_k}$, and value $V \in \mathbb{R}^{s^2\times h_v}$, where s^2 is the dimension of the input sequence, and h_k and h_v are the dimensions of the query (key) and value, respectively. Scaled dot-product attention is applied to Q, K, V

$$\mathrm{Attention}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d_k}}\right)V. \tag{6}$$

The output is obtained by computing a weighted sum of values, with the weights determined by a compatibility function between the query and the corresponding key [60]. Using different and learned projections, the query, key, and value are projected multiple times (c times), and then, these results are concatenated; this process is called MSA

$$\mathrm{MultiHead}(Q, K, V) = \mathrm{concat}(\mathrm{head}_1, \ldots, \mathrm{head}_h)W^{O} \tag{7}$$

where $\mathrm{head}_i = \mathrm{Attention}(QW^{Q}_{i}, KW^{K}_{i}, VW^{V}_{i})$, W^{Q}_{i}, $W^{K}_{i} \in \mathbb{R}^{s^2\times h_k}$, $W^{V}_{i} \in \mathbb{R}^{s^2\times h_v}$, and $W^{O} \in \mathbb{R}^{c\times h_v\times s^2}$ are learnable parameter matrices.

3) Decoder: The decoder (T_{de}) is similar to the encoder, but unlike the symmetrical structure of the autoencoder, the asymmetric encoder and decoder tend to have better results in SSL [58], [61]. The input to the decoder is the latent representation with padding of the masked tokens, which refers to adding the sequence to the position of the original masked data location, to facilitate data reconstruction through the decoder

$$\boldsymbol{G}' = \mathrm{Padding}(\boldsymbol{G}, \boldsymbol{P}_{\mathrm{m}}) \tag{8}$$

where $\boldsymbol{G}' \in \mathbb{R}^{(h+z)\times s^2}$ is the padded data. Position encodings are also added to the padding data to preserve the position information, resulting in $\boldsymbol{G}'_0$

$$\boldsymbol{G}'_0 \leftarrow \boldsymbol{G}' + \boldsymbol{E}' \tag{9}$$

where $\boldsymbol{E}' \in \mathbb{R}^{(h+z)\times s^2}$ is the position encoding information. After the position encodings and embedding operation, it is input into the transformer-based decoder for data reconstruction

$$\boldsymbol{Z} = T_{\mathrm{de}}(\boldsymbol{G}'_0) \tag{10}$$

where $\boldsymbol{Z} \in \mathbb{R}^{(h+z)\times s^2}$ is the reconstructed data. The specific operation is similar to the encoder

$$\boldsymbol{G}''_k = \boldsymbol{G}'_{k-1} + \mathrm{MSA}(\mathrm{LN}(\boldsymbol{G}'_{k-1})) \tag{11}$$

$$\boldsymbol{Z} = \boldsymbol{G}''_k + \mathrm{MLP}(\mathrm{LN}(\boldsymbol{G}''_k)) \tag{12}$$

where $k = 1, 2, \ldots, K$ represents the kth transformer layer, K is the total number of transformer layers, and $\boldsymbol{G}''_k \in \mathbb{R}^{(h+z)\times s^2}$

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is the intermediate output of the kth transformer layer. The mean squared error (MSE) between the mask data and the reconstructed mask data is calculated as the loss function

$$\text{loss}_p = \text{MSE}(\boldsymbol{P}_\text{m}, \boldsymbol{Z}_\text{m}) \tag{13}$$

where $\boldsymbol{Z}_\text{m} \in \mathbb{R}^{z\times s^2}$ is the reconstructed mask data, obtained from $\boldsymbol{Z}$ at the same position as $\boldsymbol{P}_\text{m}$.

The self-supervised pretraining process is given in Algorithm 1.

Algorithm 1 Self-Supervised Pre-Training of SSLSM

Input: The unlabeled pre-training data $\boldsymbol{X} \in \mathbb{R}^{m\times n\times l}$, the batch size bt, the number of pre-training epochs ε.
Output: The pre-training loss.
1: Obtain the $\boldsymbol{X}_{pca}$ after PCA tranform.
2: Create all sample patches in the $\boldsymbol{X}_{pca}$, and divide them into training dataset and test dataset.
3: Generate training loader.
4: **for** $i = 1$ to ε **do**
5: **for** bt samples of each batch **do**
6: Perform mask and unfold operations to get $\boldsymbol{P}_{u_m}$, $\boldsymbol{P}_m$ via equation (1).
7: Perform position encodings operation to get $\boldsymbol{P}^0_{u_m}$ via equation (2).
8: Perform the encoder (T_{en}) operation to obtain the latent representation $\boldsymbol{G}$ via equation (3).
9: Perform $Padding$ operation to get $\boldsymbol{G}'$ via equation (8).
10: Perform position encodings operation to get $\boldsymbol{G}'_0$via equation (9).
11: Perform the decoder (T_{de}) operation to obtain the reconstruction result $\boldsymbol{Z}$ via equation (10).
12: Get the data $\boldsymbol{Z}_m$ from $\boldsymbol{Z}$ at the same position as $\boldsymbol{P}_m$.
13: Perform MSE loss between the masked data $\boldsymbol{P}_m$ and the reconstructed masked data $\boldsymbol{Z}_m$ via equation (13).
14: **end for**
15: $i = i + 1$.
16: **end for**
Save the pre-trained model.

C. Model Fine-Tuning

On completion of self-supervised pretraining, the model has learned the feature representations of data. A small amount of reference samples is used for fine-tuning. Unlike the pretraining process, the input data are directly input into the encoder without spectral masking, to avoid information loss caused by spectral masking. Additionally, unlike other SSL methods based on autoencoders, the encoder and decoder are all preserved in the proposed method, which adopts a structure similar to the Unet [62] model.

First, the input $\boldsymbol{P} \in \mathbb{R}^{s\times s\times d}$ is flattened using an unfold operation. Then, position encodings $\boldsymbol{E}_\text{f} \in \mathbb{R}^{d\times s^2}$ are added to preserve spatial information. Subsequently, deep semantic features are extracted using an encoder (T_en). Similarly, position encodings $\boldsymbol{E}_\text{f} \in \mathbb{R}^{d\times s^2}$ are added to preserve spatial information. Finally, in the decoder (T_de) consisting of multiple transformer layers, deeper semantic features that the encoder cannot extract are further extracted, resulting in $\boldsymbol{F} \in R^{d\times g}$. The specific process can be formulated as follows:

$$\boldsymbol{F} = T_\text{de}(T_\text{en}(\text{Unfold}(\boldsymbol{P}) + \boldsymbol{E}_\text{f}) + \boldsymbol{E}_\text{f}). \tag{14}$$

After fine-tuning the model, the probability distribution of category information is output through LogSoftmax

$$y_\text{p} = \text{LogSoftmax}(\boldsymbol{F}) \tag{15}$$

where $\boldsymbol{F} \in \mathbb{R}^{d\times g}$ is the feature, $g = 512$, and y_p is the predicted probability distribution of each class. Finally, fine-tuning the classification task is accomplished by computing the loss between the predicted probability distribution and the true labels using the cross-entropy (CE) loss function

$$\text{loss}_\text{f} = \text{CE}(y, y_\text{p}) \tag{16}$$

where y is the ground-truth label.

The model fine-tuning process is given in Algorithm 2.

Algorithm 2 Fine-Tuning of SSLSM

Input: The $\boldsymbol{X}_{pca} \in \mathbb{R}^{m\times n\times d}$ obtained from pre-training, the ground truth label $y \in \mathbb{R}^{m\times n}$, the batch size bt, the number of pre-training epochs ε.
Output: Predicted labels of the test dataset.
1: Create all sample patches with $\boldsymbol{X}_{pca}$ and y, and divide them into training dataset and test dataset.
2: Generate training dataloader and test dataloader.
3: Load the pre-trained model.
4: **for** $i = 1$ to ε **do**
5: **for** bt samples of each batch **do**
6: Perform unfold, position encodings, encoder (T_{en}), and decoder (T_{de}) operations to get $\boldsymbol{F}$ via equation (14).
7: Perform the LogSoftmax operation to obtain the probability distribution of each class, y_p via equation (15).
8: Perform cross-entropy loss function operation via equation (16).
9: **end for**
10: $i = i + 1$.
11: **end for**
Use test dataset with the trained model to get predicted labels.

III. Experiment and Analysis

In this section, we conducted extensive experiments to evaluate the performance of the proposed SSLSM method. We first briefly describe the HSI datasets used in this study, and then, the parameters are analyzed to determine the optimal parameters. Finally, we compared the classification performance of SSLSM as implemented on the PyTorch platform and the other nine methods implemented in their original environment. The hardware platform of the experiment is Inter Xeon Gold 6130 Processor (2.1 GHz), 128-GB memory, NVIDIA GeForce RTX 3080 Ti graphics card (11 GB of RAM) workstation. To ensure the reliability of the results, each experiment was repeated five times, and the overall classification accuracy (OCA), average classification accuracy (ACA), Kappa coefficient (Kappa), and training time were used to quantify the effectiveness of methods.

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TABLE I
NUMBER OF TRAINING AND TEST SAMPLES IN INDIAN PINES, PAVIA UNIVERSITY, AND YANCHENG WETLANDS DATASETS

Class	Indian Pines			Pavia University			YC		
	Class name	Training.	Test.	Class name	Training.	Test.	Class name	Training.	Test.
C01	Alfalfa	2	44	Asphalt	53	6578	Building	18	433
C02	Corn-notill	71	1357	Meadows	149	18500	Suaeda	11	258
C03	Corn-mintill	42	788	Gravel	17	2082	River	9	205
C04	Corn	12	225	Trees	25	3039	Beach	13	318
C05	Grass-pasture	24	459	Metal Sheets	11	1334	Bare land	9	226
C06	Grass-tree	37	693	Bare soil	40	4989	Paddy field	121	2905
C07	Grass-pasture-mowed	1	27	Bitumen	11	1319	Idle land	26	624
C08	Hay-windrowed	24	454	Bricks	29	3653	Fish pond	37	883
C09	Oats	1	19	Shadows	8	939	Sea	83	1993
C10	Soybean-notill	49	923				Dry land	6	139
C11	Soybean-mintill	123	2332				Salt pan	36	865
C12	Soybean-clean	30	563				Reed	10	234
C13	Wheat	10	195				Shallow sea	62	1496
C14	Woods	63	1202				Pond	3	65
C15	Buildings-Grass-Trees	19	367				Spartina alterniflora	12	294
C16	Stone-Steel-Towers	5	88				Ditch	13	315
	Total	513	9736	Total	343	42433	Total	469	11253

A. Experimental Setup

1) HSI Datasets: To verify the effectiveness of SSLSM, three HSI datasets, including the Indian Pines dataset, Pavia University dataset, and Yancheng Wetlands dataset were used.

The Indian Pines dataset was collected by the AVIRIS in 1992. It contains 224 bands and its size is 145 × 145, with a spatial resolution of 20 m and the wavelength range of 400–2500 nm. Due to the influence of water absorption and noise, 24 bands were removed (104–108, 150–163, and 220), and the remaining 200 bands are used in this article. The false-color and ground-truth maps are shown in Fig. 2(a) and (b), respectively.

The Pavia University dataset was obtained by the ROSIS sensor with a spatial size of 610 × 340 and a spatial resolution of 1.3 m. After the noisy bands were removed, 103 bands covering the spectral range of 430–860 nm. Fig. 3(a) and (b) shows the false-color and ground-truth maps, respectively.

The Yancheng Wetlands dataset was acquired by the ZY1-02D AHSI sensor in the coastal area of Yancheng, Jiangsu, China, on September 6, 2020. The spatial size is 1398 × 942, with a spatial resolution of 30 m, covered the spectral range of 400–2500 nm. The 19 bad bands (22–27, 49–57, and 87–90) were removed, leaving 147 bands in the dataset. The false-color and ground-truth maps are shown in Fig. 4(a) and (b), respectively.

Each dataset is divided into the training set and the test set. The training sets of Indian Pines, Pavia University, and Yancheng Wetlands datasets accounted for 5%, 0.8%, and 4% of the total samples, respectively. Table I shows the land cover class names and the corresponding number of training and test samples of the three datasets.

2) Comparison of Methods: In order to validate the effectiveness of the proposed method, nine methods were selected for comparison, including K-nearest neighbor (KNN) [63], RF [64], hybrid spectral CNN (HybridSN) [26], PCA-vision transformer (PCA-ViT), semisupervised graph prototypical network (SSGPN) [65], 3-D autoencoder and Siamese network (3DAES) [66], deep few-shot learning (DFSL) [41], deep multiview learning (DMVL) [67], and unsupervised spectrum motion feature learning (SMF-UL) [68]. Among them, KNN, RF, HybridSN, and PCA-ViT only use labeled samples, and SSGPN, 3DAES, DFSL, DMVL, and SMF-UL use not only labeled samples but also unlabeled samples.

3) Parameter Setting: In the pretraining and fine-tuning stages, the number of bands is first reduced to 30 by PCA [69], and the optimizer is Adam, with a learning rate of 3e-4. The batch size is set to 128, and the number of epochs for pretraining and fine-tuning is 200 and 100, respectively. The embedding dimension in the transformer model is set to 512.

B. Parameter Sensitivity Analysis

1) Patch Size: The patch size affects the local receptive field of the model's feature learning and thus affects the performance of classification. OCA and training time with different patch sizes for three HSI datasets are shown in Fig. 5. The higher the spatial resolution of the image, the greater the purity of the pixels, and the smaller the influence of mixed pixels. For the Pavia University and Indian Pines datasets, within a certain range, the larger the patch, the higher the classification accuracy. For the Yancheng dataset, as the patch size increases, the classification accuracy declines [Fig. 5(c)]. This may be due to the mixed spatial distribution of wetlands communities and the low spatial resolution of the ZY1-02D. Increasing patch size will further increase the irrelevant redundant information, which will further be detrimental to the classification performance. In addition, the increase in patch size has a greater demand for computation, so the training time is on the rise. According to the experimental results, the appropriate patch size for the three datasets is 17, 17, and 3, respectively.

2) Mask Ratio: The ratio of spectral masking refers to the proportion of spectral bands that are masked during pretraining. With other parameters fixed, the masking ratio is set from 0% to 100% with an interval of 10%. The classification accuracy increases with the increase of the mask ratio, reaching its maximum value at 50%, and then decreases

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Fig. 2. (a) False-color and (b) ground-truth map of Indian Pines dataset.

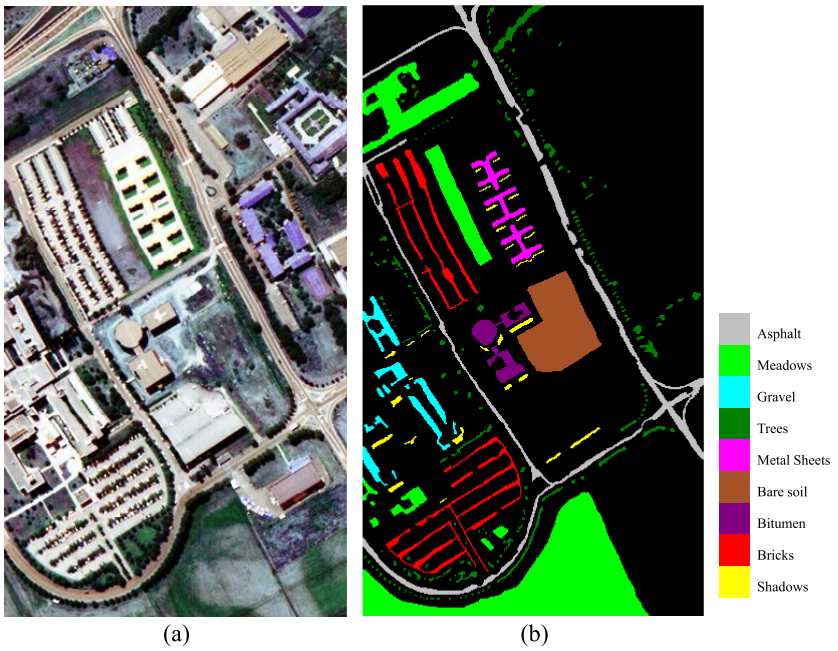

Fig. 3. (a) False-color and (b) ground-truth map of Pavia University dataset.

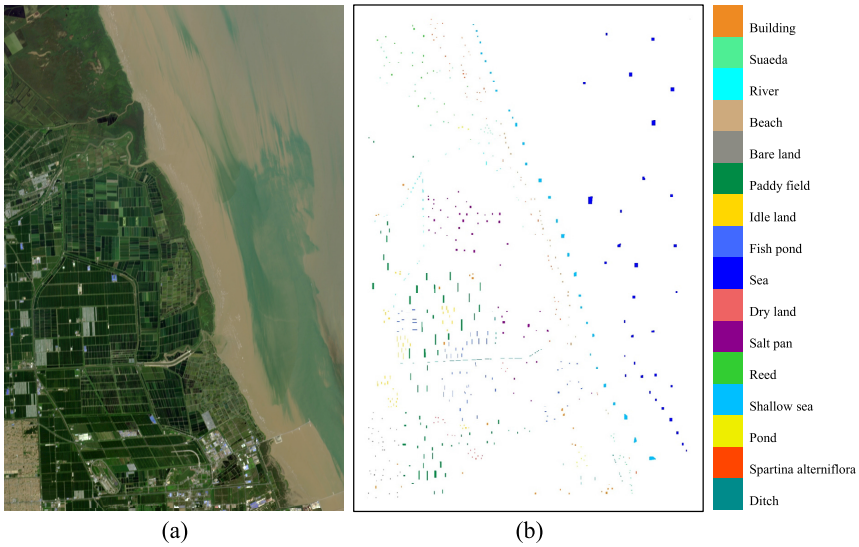

Fig. 4. (a) False-color and (b) ground-truth map of Yancheng Wetlands dataset.

as the ratio continues to increase (Fig. 6). A smaller masking ratio indicates that more spectral bands are involved, and thus, the reconstruction task is easier. However, the feature extraction ability of the pretraining model is not fully trained,

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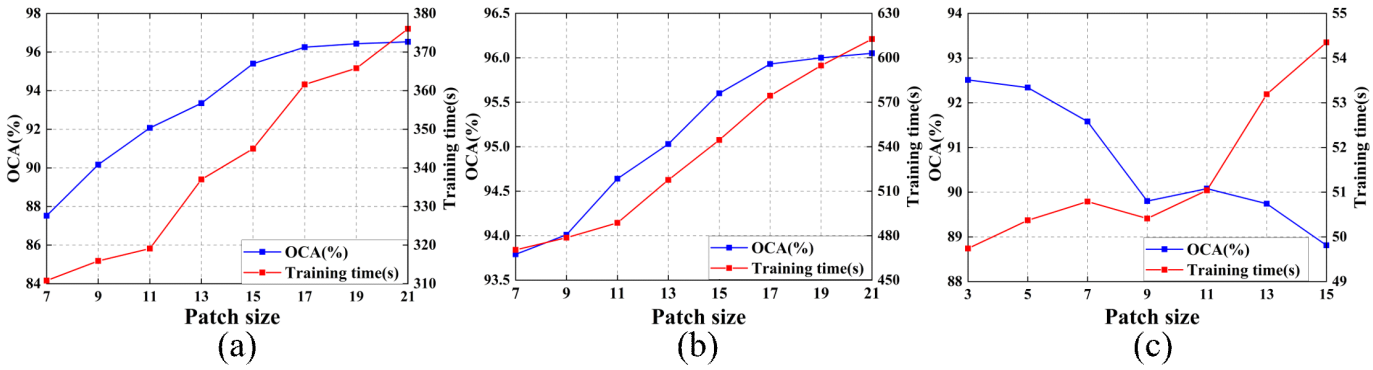

Fig. 5. Effect of patch size on OCA for: (a) Indian Pines; (b) Pavia University; and (c) Yancheng Wetlands.

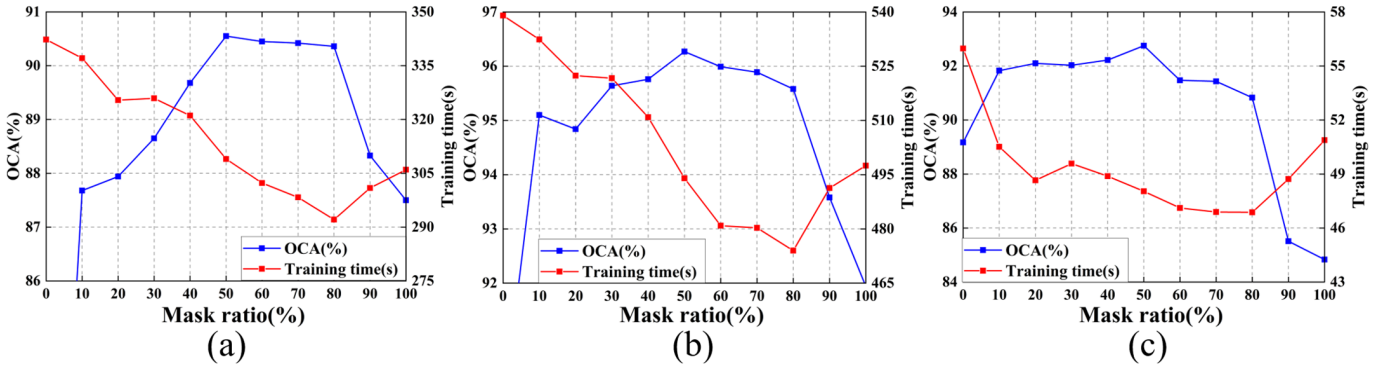

Fig. 6. Effect of mask ratio on OCA for: (a) Indian Pines; (b) Pavia University; and (c) Yancheng Wetlands.

so it results in low classification accuracy. On the other hand, when the masking ratio exceeds a certain limit, the fewer features the model can extract, resulting in information loss and therefore lower classification accuracy. When the masking ratio is less than 50%, the OCA change of the Indian Pines and Pavia University datasets is more obvious than that of Yancheng Wetlands, which indicates that the first two datasets are more sensitive to mask ratio. Overall, for the three datasets, a masking ratio of 50% is an optimal choice.

3) Model Depth: The model depth in the SSLSM is composed of the encoder and decoder depth. The encoder and decoder control the degree of feature learning, so they play an important role in the classification results. Fig. 7 shows the effect of different encoder and decoder depths on the OCA for candidate depths of {2,4,6,8} and {2,3,4}. The asymmetric encoder–decoder structure has greater advantages and is more suitable for HSI classification. Specifically, the same dataset has little performance difference at different model depths, which may be caused by using PCA to reduce the dimension to the same dimension, and the difference in spectral reconstruction is not large. In terms of the OCA of three datasets, the reasonable model structure is {8,2}, {2,4}, and {2,4}, respectively.

4) Sample Percentage for Pretraining: During the pretraining phase, the size of samples was determined according to the percentage of labeled pixels in the HSI. The impacts of sample percentage on OCA are given in Fig. 8. The results show that, first, the more the self-supervised samples, the more comprehensive the features the model learns, and the higher the classification accuracy. However, as the sample percentage further increases, the OCA almost no longer increases. Different from OCA, the training time of the three datasets increases linearly with the increase of the sample percentage. Taking into account both classification accuracy and training time, the percentage of training samples of 50% (OCA: 90.49%), 35% (OCA: 96.32%), and 20% (OCA: 93.89%) are used for the three datasets, respectively.

5) Sample Percentage for Fine-Tuning: In the model fine-tuning phase, ground-truth labels are used to fine-tune the model parameters. Similar to the pretraining stage, when the sample size increases, OCA shows an increasing trend, while training time also increases (Fig. 9). However, the increase rate of OCA slows down after the sample percentage reaches a certain value. The proposed method can achieve high classification accuracy even with a small percentage of samples, indicating that the model can effectively extract land cover features. The percentage of fine-tuning samples for the three datasets is 5%, 0.8%, and 4%, respectively.

C. Classification Results

1) Classification Accuracy: Tables II–IV represent the classification accuracies of ten methods on the Indian Pines, Pavia University, and Yancheng Wetlands datasets, respectively. The results show that the proposed SSLSM method achieved the optimal ACA, OCA, and Kappa. SSLSM can extract HSI features by training the model through the pretext task of spectral masking, and further feature extraction is carried out using the decoder during the fine-tuning stage. In terms of classification performance on the three datasets,

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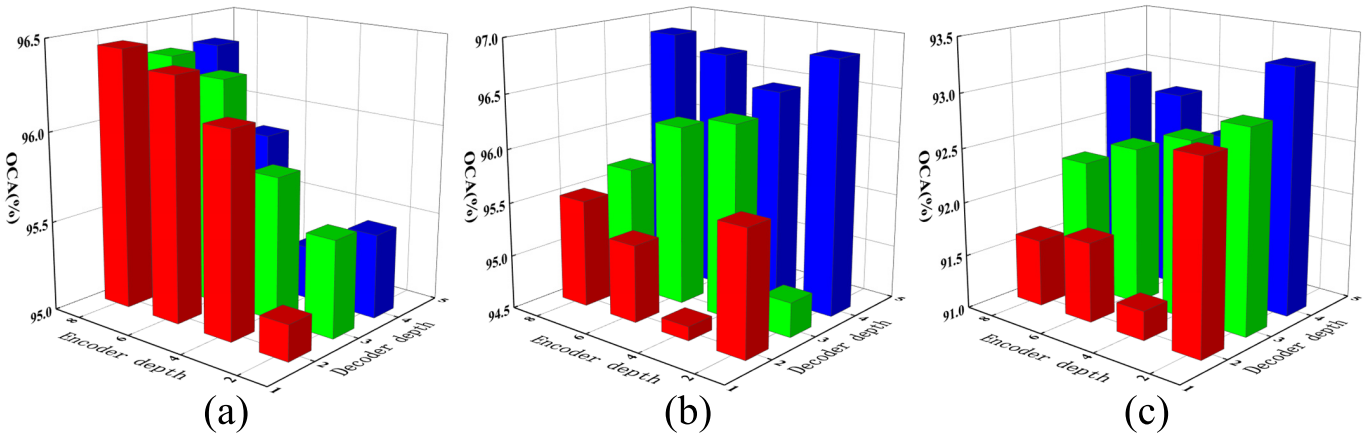

Fig. 7. Effect of encoder and decoder depth on OCA for: (a) Indian Pines; (b) Pavia University; and (c) Yancheng Wetlands.

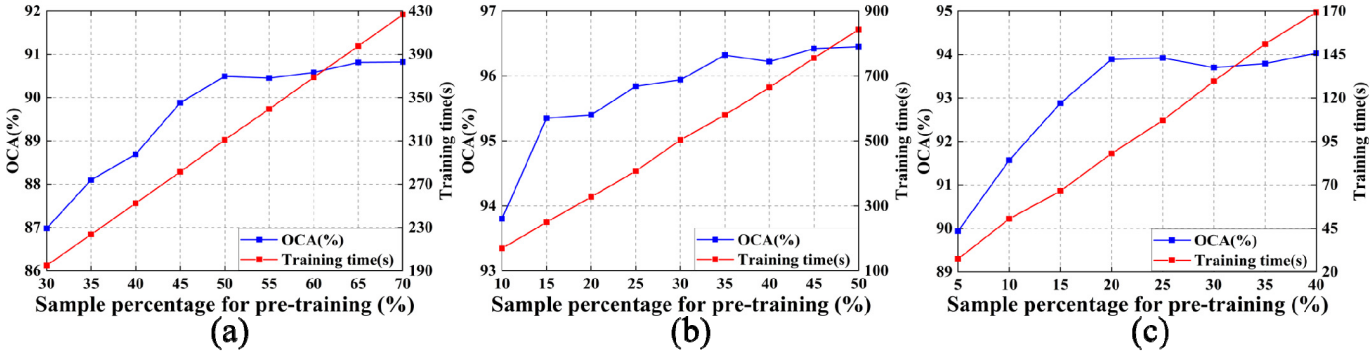

Fig. 8. Effect of sample percentages of pretraining on OCA for: (a) Indian Pines; (b) Pavia University; and (c) Yancheng Wetlands.

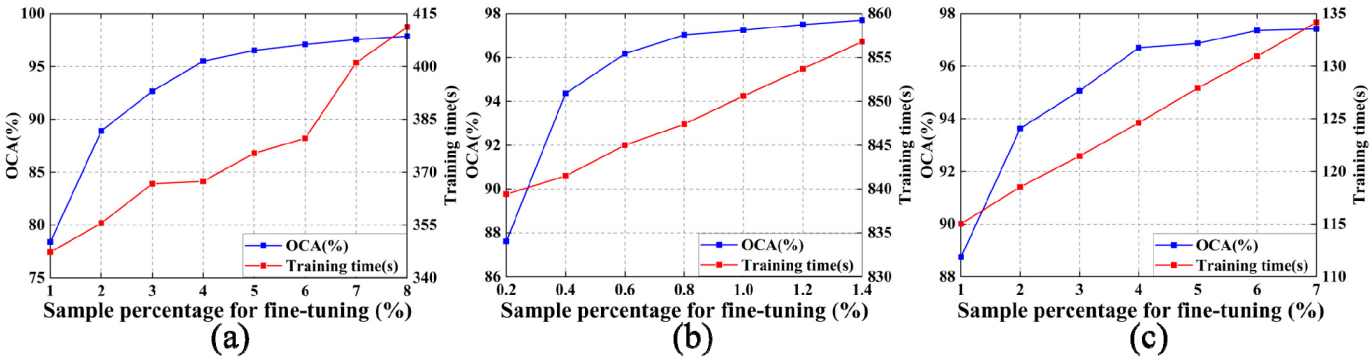

Fig. 9. Effect of fine-tuning sample percentages on OCA for: (a) Indian Pines; (b) Pavia University; and (c) Yancheng Wetlands.

the SMF-UL method is suboptimal, with Kappa of 94.25%, 93.45%, and 94.34%, respectively.

For the Grass Pasture Mowed and Oats in the Indian Pines dataset, both RF and PCA-ViT show relatively low classification accuracy. This could be due to the limited labeled samples for these classes, resulting in fewer samples being used for training and thus lower classification accuracy. However, even with enough training samples, traditional methods like KNN and RF exhibit relatively low accuracy, such as in the case of Gravel and Bare Soil in the Pavia University dataset. This may be due to the similar spectra of these classes or the large spectral heterogeneity of the same object.

In general, some methods such as 3DAES, DMVL, and SMF-UL that do not require a large number of labeled samples can make full use of both labeled and unlabeled samples and avoid overfitting during the training process, resulting in higher classification accuracy. However, the results in Table IV demonstrate that both the DFSL and DMVL methods did not achieve satisfactory results and their accuracy was even lower than that of traditional methods. This may be due to the influence of mixed pixels in low-resolution HSI, because it is difficult to get pure pixels of a certain ground object, thus reducing the accuracy of feature representation learned by the pretraining.

The proposed method exhibits relatively consistent and demonstrates the effectiveness of the model in small sample scenarios. It presents a feasible solution for SSL of HSI.

2) Classification Maps: Figs. 10–12 show the mapping results of ten methods for Indian Pines, Pavia University,

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TABLE II

CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS ON THE INDIAN PINES DATASET (THE OPTIMAL RESULT IS HIGHLIGHTED IN BOLD FONT)

	KNN	RF	HybridSN	PCA-ViT	SSGPN	3DAES	DFSL	DMVL	SMF-UL	SSLSM
C01	33.18±11.19	2.72±2.65	55.46±12.61	24.54±7.44	36.36±51.22	71.36±22.39	41.82±14.42	81.14±12.57	85.00±16.43	**98.18±1.90**
C02	49.90±2.15	56.48±3.62	91.72±2.29	76.57±1.49	92.53±2.44	89.68±1.64	80.13±2.95	93.10±3.27	92.00±0.92	**93.62±1.23**
C03	42.94±2.55	46.84±4.25	82.51±8.77	81.80±4.75	89.19±2.06	95.31±2.90	76.95±4.87	92.40±2.70	95.49±2.02	**97.72±0.91**
C04	32.98±3.83	27.38±11.04	76.36±9.70	54.76±7.51	78.52±15.18	78.04±5.44	56.93±5.28	95.24±3.21	88.98±6.91	**95.38±1.02**
C05	77.12±2.24	78.69±3.80	92.16±3.71	95.64±0.95	93.83±2.14	85.84±3.61	84.23±5.79	89.02±6.04	90.46±3.49	**99.13±0.99**
C06	90.94±2.43	92.76±3.37	94.37±2.52	92.09±2.86	**99.38±0.96**	97.75±0.96	88.20±4.34	91.27±5.51	99.11±0.51	98.47±6.73
C07	63.70±25.60	0.74±1.48	71.85±20.72	7.41±4.53	33.33±54.56	60.00±27.66	39.63±13.36	36.30±40.77	46.67±14.17	**86.67±1.15**
C08	95.02±2.50	97.75±1.42	98.37±2.34	98.64±1.16	99.85±0.25	**100±0**	86.72±2.71	99.91±0.15	**100±0**	98.19±2.88
C09	13.68±12.28	5.26±10.53	78.95±17.06	20.00±18.00	45.61±48.90	54.74±25.70	31.05±14.78	54.21±24.35	27.37±12.19	**97.89±1.64**
C10	57.83±3.65	61.26±4.21	93.76±2.58	79.98±2.40	86.64±6.66	86.78±1.47	73.67±3.53	90.81±3.24	87.87±2.91	**95.54±0.68**
C11	67.52±1.68	82.81±2.75	95.94±0.95	92.68±1.20	95.54±0.78	96.78±1.25	88.57±2.63	95.56±1.21	97.23±1.86	**97.95±2.08**
C12	36.06±1.74	40.50±3.75	76.06±7.41	59.18±4.95	80.52±14.99	86.18±5.33	68.19±6.71	88.65±2.64	92.58±1.83	**93.07±0.56**
C13	89.23±4.53	91.79±3.52	87.18±12.35	89.85±3.69	99.83±0.29	99.49±0.46	87.85±5.55	86.92±7.37	**99.90±0.21**	99.59±0.25
C14	89.85±3.66	92.55±2.10	97.72±0.93	97.10±0.83	**99.61±0.39**	98.32±0.60	93.25±1.73	95.43±1.95	99.05±0.40	99.35±1.30
C15	28.28±3.47	31.28±7.24	72.81±7.09	82.07±2.38	85.92±10.59	98.37±1.26	71.88±3.67	93.90±2.68	**98.96±0.47**	91.23±4.75
C16	82.95±1.25	65.91±15.61	78.18±6.60	68.64±10.89	**99.62±0.66**	95.68±1.33	42.16±7.73	53.30±24.92	96.59±1.44	69.32±0.89
ACA	59.45±0.83	54.67±1.62	83.96±3.41	70.06±0.69	82.27±1.96	87.15±2.66	69.45±1.48	83.57±3.75	87.33±1.71	**94.46±0.89**
Kappa	59.43±0.69	65.55±0.47	89.71±1.84	82.92±0.97	91.63±1.21	92.44±0.41	79.23±1.02	91.73±0.52	94.25±0.89	**96.03±0.46**
OCA	64.52±0.57	70.21±0.33	91.01±1.60	85.10±0.86	92.62±1.14	93.38±0.36	81.87±0.88	92.75±0.45	94.96±0.78	**96.52±0.40**

TABLE III

CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATASET (THE OPTIMAL RESULT IS HIGHLIGHTED IN BOLD FONT)

	KNN	RF	HybridSN	PCA-ViT	SSGPN	3DAES	DFSL	DMVL	SMF-UL	SSLSM
C01	80.50±2.99	83.26±3.41	87.72±3.31	87.44±1.17	96.72±0.96	96.91±1.01	78.99±3.36	88.59±3.66	97.46±1.32	**98.86±2.17**
C02	90.48±1.43	94.71±1.19	99.77±0.20	99.61±0.46	99.47±0.26	98.24±0.82	97.11±0.98	97.90±0.94	98.71±0.69	**99.82±1.36**
C03	45.98±3.99	31.26±4.56	60.18±9.07	77.23±1.83	75.82±7.56	85.29±2.81	56.83±5.29	**91.71±4.80**	86.38±3.43	85.98±4.29
C04	71.59±5.80	78.70±2.94	75.47±5.51	60.32±1.06	88.32±3.20	95.54±1.43	49.88±5.93	54.49±7.34	**97.90±0.50**	96.58±1.26
C05	98.65±0.98	98.14±0.61	95.55±3.40	99.51±0.24	**100±0**	99.67±0.08	88.30±6.58	98.76±3.54	99.71±0.21	99.07±0.71
C06	49.47±2.29	37.99±3.54	98.26±1.80	88.96±2.17	86.47±2.61	78.46±3.42	94.04±1.33	94.63±1.11	83.28±2.74	**98.44±1.24**
C07	75.50±5.19	62.50±8.48	90.37±11.73	**98.88±0.46**	74.19±15.71	78.46±9.44	73.72±9.07	94.32±4.05	77.52±4.64	93.18±6.62
C08	75.77±2.97	82.43±2.42	82.07±3.61	65.72±4.14	88.04±4.99	94.45±2.45	67.96±3.30	93.56±5.15	**94.89±3.09**	91.71±0.72
C09	99.74±0.13	99.08±0.52	61.73±12.59	33.06±7.19	**100±0**	99.89±0.12	71.65±8.09	28.30±8.03	99.32±0.68	91.95±2.58
ACA	76.41±0.97	74.23±1.46	83.46±3.01	78.97±0.50	89.89±2.16	91.88±0.97	75.39±2.18	82.47±1.31	92.80±0.61	**94.73±1.03**
Kappa	72.17±0.53	72.95±0.91	88.32±1.77	84.06±0.44	91.72±1.38	92.00±0.57	79.13±1.55	87.60±1.14	93.45±0.90	**96.06±0.69**
OCA	79.31±0.38	80.15±0.66	91.25±1.32	88.14±0.33	93.81±1.02	94.02±0.43	84.50±1.15	90.66±0.88	95.10±0.67	**97.03±0.52**

TABLE IV

CLASSIFICATION ACCURACY (%) OF DIFFERENT METHODS ON THE YANCHENG WETLANDS DATASET (THE OPTIMAL RESULT IS HIGHLIGHTED IN BOLD FONT)

	KNN	RF	HybridSN	PCA-ViT	SSGPN	3DAES	DFSL	DMVL	SMF-UL	SSLSM
C01	81.34±6.13	85.87±10.39	90.76±3.56	82.49±12.75	**99.23±1.14**	95.10±1.22	70.18±6.46	67.23±7.53	96.08±1.20	95.62±3.19
C02	94.34±2.31	**96.90±1.45**	69.69±9.84	67.67±7.13	88.50±7.34	89.92±3.86	68.37±5.50	61.20±8.57	91.27±6.51	84.19±4.43
C03	81.94±5.13	76.60±6.20	85.53±7.12	76.50±10.88	87.05±9.96	93.62±4.64	52.73±6.61	81.66±11.30	**96.21±2.41**	94.27±1.39
C04	87.30±10.27	85.22±3.30	**89.75±4.43**	55.79±17.85	26.94±32.84	76.39±11.15	66.35±5.38	89.28±3.37	74.98±10.23	86.60±4.20
C05	99.65±0.43	99.56±0.89	**99.91±0.20**	87.70±24.11	99.56±0	98.42±0.71	86.11±5.64	81.11±11.85	97.79±0.56	99.44±0.20
C06	98.03±1.18	99.70±0.59	98.71±0.66	98.70±1.01	99.16±1.45	99.68±0.49	96.63±1.05	96.13±1.29	99.59±0.62	**99.71±0.18**
C07	99.62±0.69	95.83±3.40	97.95±1.88	96.99±3.90	97.54±3.21	99.79±0.14	88.70±2.00	79.97±7.81	98.69±1.84	**99.85±0.29**
C08	78.87±2.92	79.23±2.36	**92.37±2.84**	83.71±2.87	85.01±3.43	85.86±3.95	77.06±3.75	86.32±2.86	87.67±1.17	87.72±1.25
C09	**100±0**	99.86±0.06	99.90±0.22	99.48±1.11	99.97±0.03	**100±0**	96.46±1.64	99.81±0.31	**100±0**	**100±0**
C10	71.94±6.44	89.78±8.60	87.89±12.90	60.57±32.58	76.98±14.69	66.43±15.19	78.27±12.44	52.59±11.22	69.57±1.40	**92.52±8.70**
C11	95.21±1.49	96.76±0.71	92.35±2.42	85.83±13.48	**97.99±1.47**	95.05±2.19	77.99±4.80	91.72±1.25	96.13±0.51	97.87±0.96
C12	96.15±2.18	**98.29±1.64**	92.31±4.70	87.61±9.10	95.59±3.21	93.73±2.31	87.99±4.39	67.74±14.44	93.02±2.36	94.79±1.22
C13	99.69±0.33	**99.87±0.13**	94.91±9.89	97.83±2.89	99.73±0.46	97.64±2.90	91.16±2.78	99.49±1.16	99.69±0.33	99.79±0.27
C14	47.08±11.94	34.15±12.81	**86.15±8.36**	61.23±4.27	51.28±10.47	16.67±20.42	17.85±5.25	77.85±9.66	35.15±7.13	72.00±11.33
C15	84.83±3.26	97.76±1.85	96.19±3.24	87.01±12.01	91.50±7.49	96.22±3.99	51.80±6.61	38.10±9.39	98.51±1.02	**99.52±0.57**
C16	53.97±7.08	59.94±9.32	74.28±8.70	64.83±5.05	61.16±12.56	54.21±12.33	62.48±7.77	65.08±4.75	54.30±5.62	**76.83±2.51**
ACA	86.52±1.64	87.21±1.57	90.35±1.05	80.87±8.04	84.83±3.02	84.92±1.49	73.13±1.59	77.20±1.20	86.79±1.02	**92.54±0.85**
Kappa	92.40±0.58	93.71±0.58	94.00±1.42	89.68±4.50	92.68±0.83	93.57±0.26	83.56±0.92	87.15±0.40	94.34±0.56	**96.18±0.28**
OCA	93.44±0.50	94.58±0.50	94.82±1.24	91.12±3.84	93.71±0.70	94.46±0.22	85.95±0.79	88.93±0.35	95.12±0.48	**96.70±0.24**

and Yancheng Wetlands datasets, respectively. Compared with other methods, the mapping results of SSLSM are most like the ground truth. Methods such as KNN, RF, and PCA-ViT clearly have significant noise and a high degree of misclassification. This is consistent with the classification accuracy shown in Tables II–IV. For Indian Pines, the Soybean-mintill in the ellipse is easily misclassified as Corn-notill by other methods, while SSLSM can identify it more accurately, and similar results can be found in Figs. 11 and 12. Additionally, the mapping results of SSLSM have clear contours and obvious boundaries between ground objects, which also verifies the effectiveness of our method. For the

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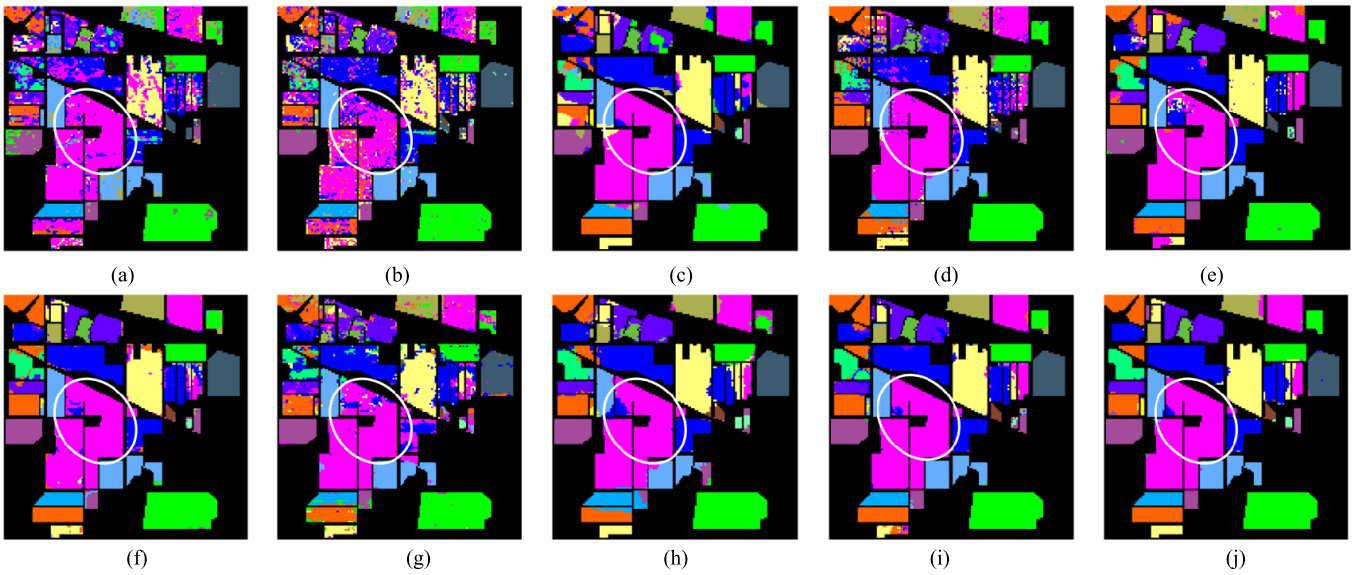

Fig. 10. Classification results of different methods applied to the Indian Pines dataset with the corresponding OCA: (a) KNN, OCA = 64.52%; (b) RF, OCA = 70.21%; (c) HybridSN, OCA = 91.01%; (d) PCA-ViT, OCA = 85.10%; (e) SSGPN, OCA = 92.62%; (f) 3DAES, OCA = 93.38%; (g) DFSL, OCA = 81.87%; (h) DMVL, OCA = 92.75%; (i) SMF-UL, OCA = 94.96%; and (j) SSLSM, OCA = 96.52%.

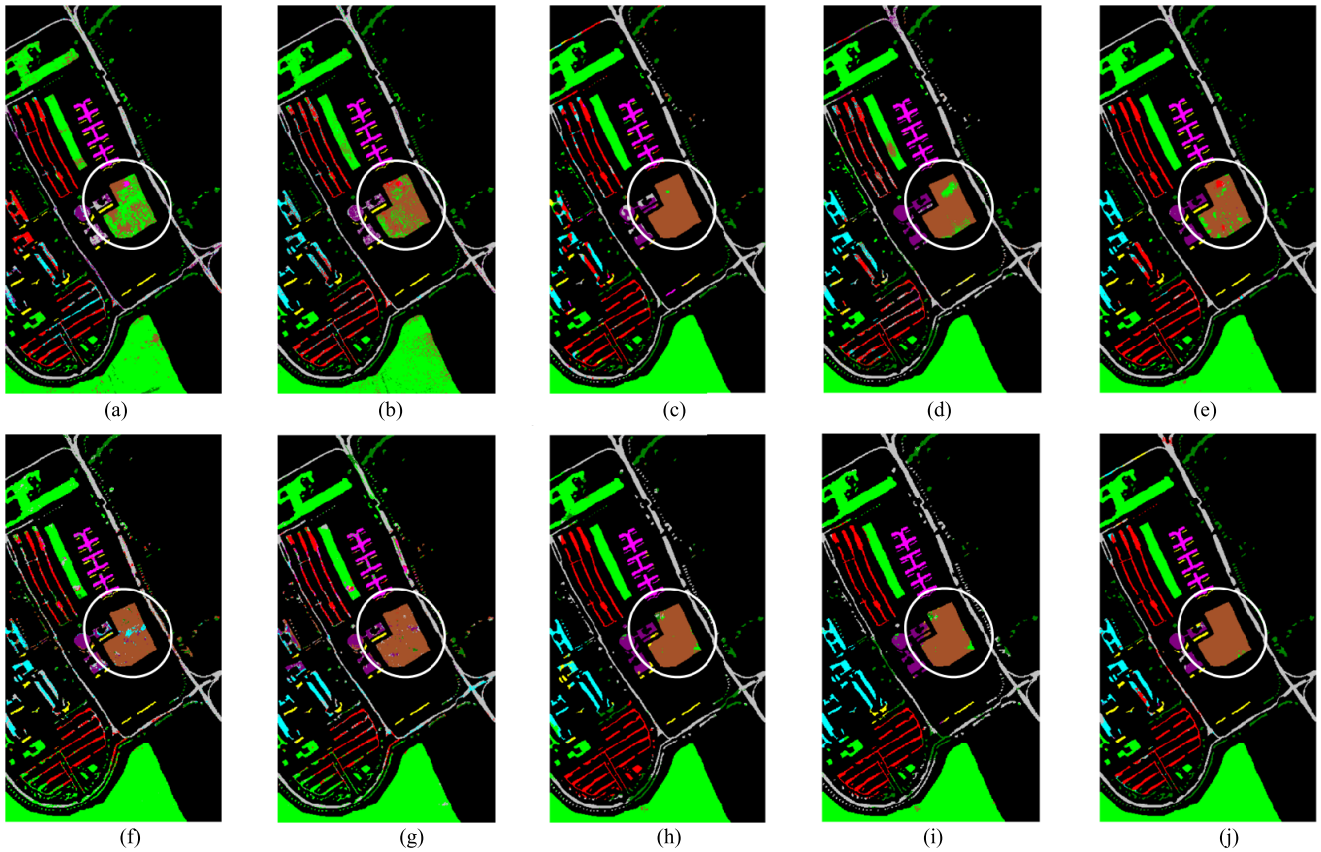

Fig. 11. Classification results of different methods applied to the Pavia University dataset with the corresponding OCA: (a) KNN, OCA = 79.31%; (b) RF, OCA = 80.15%; (c) HybridSN, OCA = 91.25%; (d) PCA-ViT, OCA = 88.14%; (e) SSGPN, OCA = 93.81%; (f) 3DAES, OCA = 94.02%; (g) DFSL, OCA = 84.50%; (h) DMVL, OCA = 90.66%; (i) SMF-UL, OCA = 95.10%; and (j) SSLSM, OCA = 97.03%.

Yancheng Wetlands dataset, SSLSM can accurately delineate the wetland vegetation such as *Spartina alterniflora* and *reed*, which can provide guidance and suggestions for policymakers in wetland restoration and protection.

3) Time Cost: The time cost is another important indicator for evaluating the effectiveness of classification methods, especially for deep learning models. Table V shows the training time for the ten methods. Since classification methods

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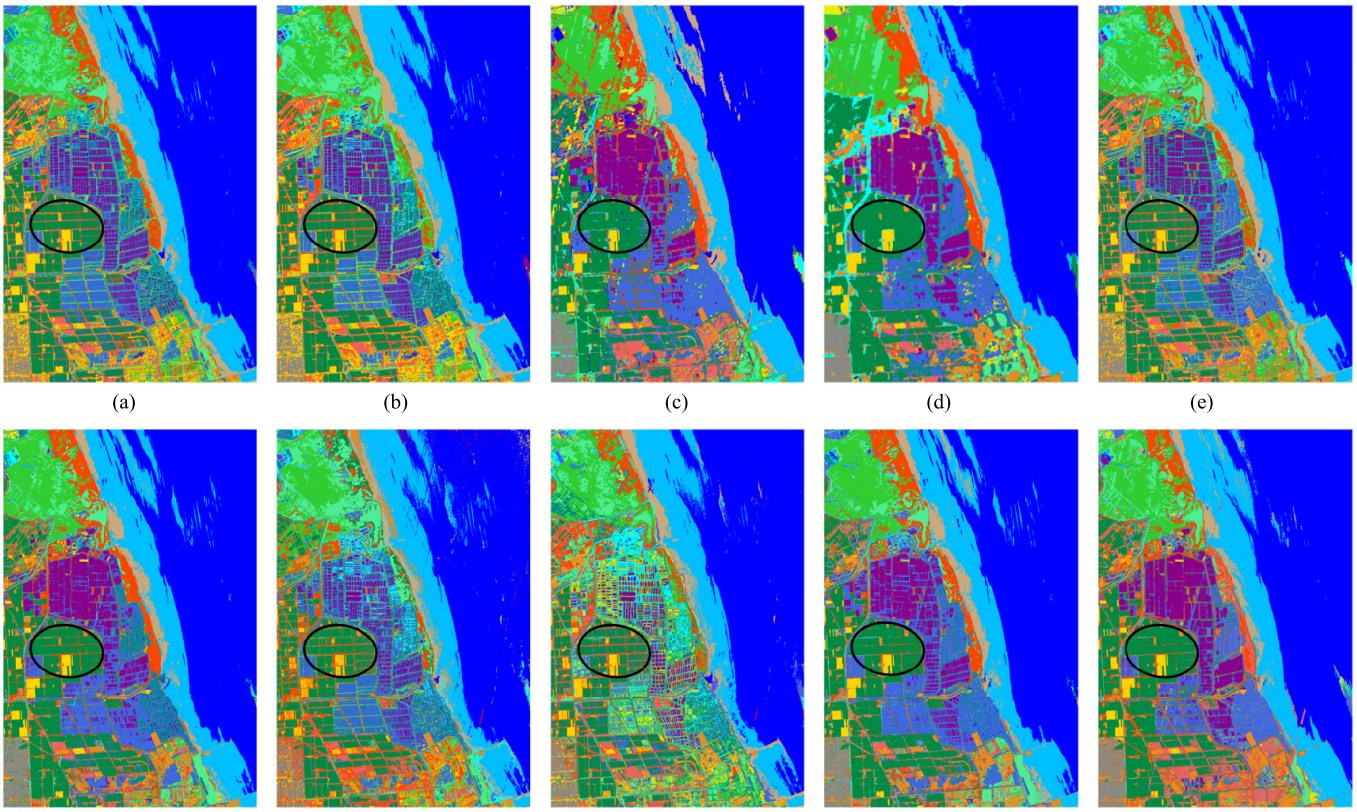

Fig. 12. Classification maps of different methods applied to the Yancheng Wetlands dataset with the corresponding OCA: (a) KNN, OCA = 93.44%; (b) RF, OCA = 94.58%; (c) HybridSN, OCA = 94.82%; (d) PCA-ViT, OCA = 91.12%; (e) SSGPN, OCA = 93.71%; (f) 3DAES, OCA = 94.46%; (g) DFSL, OCA = 85.95%; (h) DMVL, OCA = 88.93%; (i) SMF-UL, OCA = 95.12%; and (j) SSLSM, OCA = 96.70%.

TABLE V
TRAINING TIME (/S) OF DIFFERENT METHODS FOR THREE DATASETS

	KNN	RF	HybridSN	PCA-ViT	SSGPN	3DAES	DFSL	DMVL	SMF-UL	SSLSM
Indian Pines	1.21	2.94	16.84	72.70	578.92	490.68	1102.42	1678.21	645.23	375.40
Pavia University	10.32	11.34	18.21	48.99	1651.89	958.77	1088.45	9573.69	1129.32	847.41
Yancheng Wetlands	6.32	7.48	8.39	43.75	239.60	380.25	1568.97	3176.36	502.21	124.62

TABLE VI
EFFECT OF DIFFERENT FINE-TUNING MODELS ON CLASSIFICATION ACCURACY (%) OF THREE DATASETS

	Indian Pines			Pavia University			Yancheng Wetlands		
	ACA	Kappa	OCA	ACA	Kappa	OCA	ACA	Kappa	OCA
No-decoder	87.68±1.48	92.07±0.47	93.42±0.42	78.02±0.85	86.78±0.27	90.12±0.20	82.27±0.82	91.34±0.41	92.56±0.35
SSLSM	94.46±0.89	96.03±0.46	96.52±0.40	94.73±1.03	96.06±0.69	97.03±0.52	92.54±0.85	96.18±0.28	96.70±0.24

TABLE VII
EFFECT OF PRETEXT TASK ON CLASSIFICATION ACCURACY (%) OF THREE DATASETS

	Indian Pines			Pavia University			Yancheng Wetlands		
	ACA	Kappa	OCA	ACA	Kappa	OCA	ACA	Kappa	OCA
FS	71.23±1.65	76.23±0.52	80.52±0.49	75.09±2.06	79.56±0.46	84.74±0.30	77.68±1.46	86.52±0.66	89.97±0.57
SSLSM	94.46±0.89	96.03±0.46	96.52±0.40	94.73±1.03	96.06±0.69	97.03±0.52	92.54±0.85	96.18±0.28	96.70±0.24

such as HybridSN do not require a complex feature extraction process, they consume less time for training. However, the accuracy of these classification methods cannot be further improved. Models that are capable of pretraining on abundant unlabeled samples, such as 3DAES, typically employ deep neural network models to extract high-level features. These models have more parameters and require more training time. However, the proposed method adopts a two-stage strategy of

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TABLE VIII
EFFECT OF DIFFERENT TYPES OF PRETEXT TASKS ON CLASSIFICATION ACCURACY (%) OF THREE DATASETS

	Indian Pines			Pavia University			Yancheng Wetlands		
	ACA	Kappa	OCA	ACA	Kappa	OCA	ACA	Kappa	OCA
Spatial masking	90.19± 0.44	92.10± 0.14	93.08± 0.12	91.34± 0.52	91.82± 0.63	93.86± 0.47	90.61± 0.68	92.45± 0.28	93.07± 0.24
Rotate	71.54± 0.77	83.52± 0.12	85.60± 0.09	82.30± 3.39	86.93± 2.21	90.24± 1.66	85.98± 0.63	91.82± 0.44	92.72± 0.38
Spectral masking	94.46± 0.89	96.03± 0.46	96.52± 0.40	94.73± 1.03	96.06± 0.69	97.03± 0.52	92.54± 0.85	96.18± 0.28	96.70± 0.24

pretraining and fine-tuning, which can control the time cost well and ensure classification efficiency.

4) Ablation Experiment:

a) Effect of the decoder in fine-tuning on classification accuracy: During the pretext task, the encoder and decoder are used for feature learning and band reconstruction, respectively. While in the fine-tuning phase, the encoder and decoder are cascaded in SSLSM during fine-tuning. In this section, we analyzed the impact of the decoder used in the fine-tuning stage on the classification performance of HSI (Table VI). The decoder can further improve the classification performance of Kappa from 92.07%, 86.78%, and 91.34% to 96.03%, 96.06%, and 96.18% for Indian Pines, Pavia University, and Yancheng Wetlands, respectively. In addition, the classification performance is more significantly improved with a higher number of decoder layers. For example, the Indian Pines dataset has a decoder layer of 2, and the Pavia University dataset has a decoder layer of 4.

b) Effect of pretext task on classification accuracy: To evaluate the importance of the pretraining on HSI classification accuracy, a fully supervised (FS) method, that only retains the fine-tuning stage without the self-supervised pretraining, was proposed and compared with SSLSM. ACA increased from 71.23, 75.09, and 77.68 to 94.46, 94.73, and 92.54 for Indian Pines, Pavia University, and Yancheng Wetlands, respectively (Table VII). A similar significant improvement was also witnessed in Kappa and OCA. These results indicate that an effective pretraining process can effectively learn more features and thus improve classification performance.

c) Effect of different types of pretext tasks on classification accuracy: In the field of SSL, the choice of pretext task is crucial for the accuracy of downstream HSI classification. To verify the necessity of the pretext task of spectral masking, the other two pretext tasks, SM and RP, were included and compared (Table VIII). For SM, 50% of the pixels in the image patch were masked, while for RP, rotations of 90°, 180°, and 270° were selected. The results show that the RP had the lowest classification accuracy, likely due to its simplicity and inability to capture the complex features of HSI. In contrast, SM not only lost spatial information but also spectral information, making the reconstruction task more challenging. While spectral masking only lost some spectral information, making reconstruction moderately difficult and achieving the best accuracy. Therefore, the spectral masking pretext task is crucial for accurate downstream HSI classification.

IV. CONCLUSION

This article proposes a self-supervised classification method based on spectral masking for HSI, named SSLSM. The method includes two stages: pretraining and fine-tuning. In the pretraining stage, to fully utilize the rich spectral information of HSI and maintain the spatial continuity of HSI, an innovative spectral masking method is used to construct the pretext task. In the fine-tuning stage, considering that the decoder is not been fully utilized, we creatively cascade the encoder and decoder to further extract deep semantic features, and ultimately apply them to HSI classification. Our proposed method achieved optimal quantitative and qualitative classification results on the three datasets. In future research, we will explore the application of SSL methods on a larger scale HSI classification.

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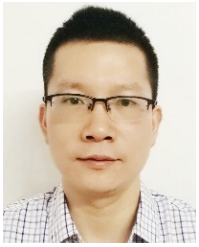

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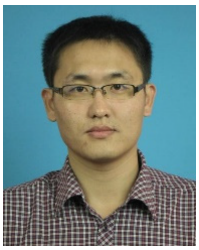

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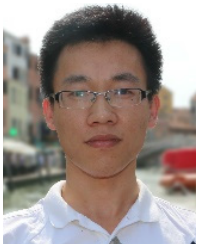

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